z-Transform Analysis of LTI Systems

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Key Concepts

1) Linear time-invariant systems described by difference equations play an important role in signal processing.

    a) They are widely used as filters.

    b) They are used as models for distortion.

    c) They are used as models for characterizing random signals.

2) If the difference equation takes the form

$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k]$$

where $a_0 \neq 0$,

a) The system function is given by

$$H(z) = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}$$

b) The system function is expressed in terms of poles d_k and zeros c_k as (assuming $b_0 \neq 0$)

$$H(z) = \frac{b_0}{a_0} \frac{\prod_{k=1}^{M} (1 - c_k z^{-1})}{\prod_{k=1}^{N} (1 - d_k z^{-1})}$$

3) Important questions:

a) What are the conditions for a system to be stable?

b) What are the conditions for a system to be causal?

c) If $H(z)$ is a model for a system that introduces distortion, can it be reversed or inverted?

d) How does the frequency response relate to the system function?

Video Screenshots

LTI systems described by difference equations play an important role in signal processing

1) Filters: infinite impulse response (IIR)
finite impulse response (FIR)

a) Filter design

b) Filter implementation

2) Models:

a) for a system introducing distortion

b) for characteristics of a random process

General form $\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k]$, $a_0 \neq 0$

$$\sum_{k=0}^{N} a_k z^{-k} Y(z) = \sum_{k=0}^{M} b_k z^{-k} X(z)$$

$$Y(z) = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}} X(z)$$

$$H(z) = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}} = \frac{b_0}{a_0} \frac{\prod_{k=1}^{M} (1 - c_k z^{-1})}{\prod_{k=1}^{N} (1 - d_k z^{-1})}$$

transfer or system function

c_k: zeros

d_k: poles

assumes $b_0 \neq 0$

Example: Given H(z), find difference equation

$$H(z) = \frac{(z+1)^2}{(z-1/2)(z+1/4)}$$

write in terms of z^{-1}

$$= \frac{z^2(1+z^{-1})^2}{z^2(1-1/2\,z^{-1})(1+1/4\,z^{-1})}$$

zeros: $z = -1$ (x 2)

poles: $z = 1/2, -1/4$

multiply terms

$$= \frac{1 + 2z^{-1} + z^{-2}}{1 - 1/4\,z^{-1} - 1/8\,z^{-2}}$$

$b_0 = 1, \; b_1 = 2, \; b_2 = 1$

$a_0 = 1, \; a_1 = -1/4, \; a_2 = -1/8$

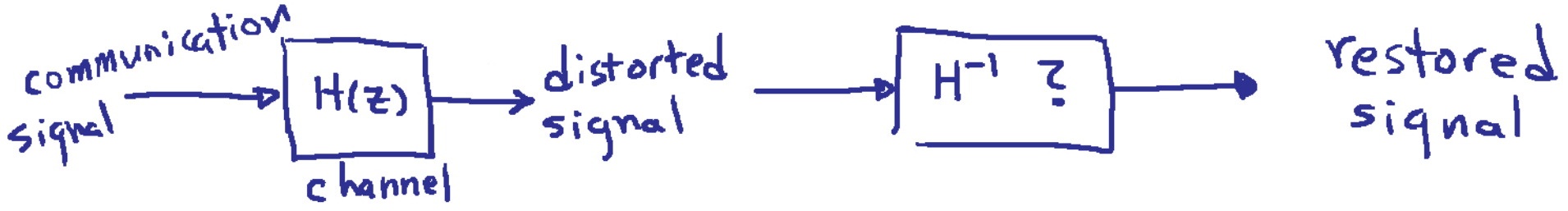
$$y[n] - 1/4\,y[n-1] - 1/8\,y[n-2] = x[n] + 2x[n-1] + x[n-2]$$

Key questions –

1) What are the conditions for a stable system?

2) What are the conditions for a causal system?

3) If $H(z)$ is a model for a system that introduces distortion, can it be reversed?



4) How does the frequency response relate to the parameters of the difference equation?

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